

DIGT 3107 SECTION A

Phase 2 Submission

Student#: 221888888

Sur (Family) Name: Arnaud

Given Name: Katherine

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Table Descriptions

Guest

The Guest table stores information about each person who makes or stays under a booking. It contains the guest's name, email address, phone number, and whether the guest is university staff. The **GuestID** attribute is the primary key and uniquely identifies each guest.

RoomType

The RoomType table stores the available categories of rooms at Elmwood University Guest House. The three room types established in Phase 1 are Single, Duplex, and Presidential Suite. Each room type contains its capacity and nightly price. **RoomTypeID** is the primary key.

Room

The Room table represents each physical room in the guest house. Each room has a unique room number and is associated with one room type through the **RoomTypeID** foreign key. **RoomID** is the primary key.

Booking

The Booking table records guest reservations. Each booking connects one guest to one room and stores the booking date/time, check-in and check-out dates, payment method, booking status, and cancellation date when applicable. **BookingID** is the primary key. **GuestID** and **RoomID** are foreign keys referencing the Guest and Room tables.

Test Data Explanation

Sample data was created to test the relationships and functional requirements of the database. The data includes university staff and non-staff guests, all three available room types, multiple physical rooms, and bookings with Active, Completed, and Cancelled statuses. Several bookings occur during 2025 so that future queries can calculate annual revenue, monthly revenue, room popularity, and cancellation trends. Cancelled bookings include a cancellation date, while active and completed bookings leave this field null.

Screenshots of Tables

Booking table data

BookingID	GuestID	RoomID	BookingDateTime	CheckInDateTime	CheckOutDateTime	PaymentMethod	Status	CancellationDateTi...
1	1	1	2025-01-03 09:15:00	2025-01-10 15:00:00	2025-01-12 11:00:00	Credit Card	Completed	NULL
2	2	4	2025-02-05 14:20:00	2025-02-20 15:00:00	2025-02-23 11:00:00	Debit Card	Completed	NULL
3	3	7	2025-03-11 10:00:00	2025-03-25 15:00:00	2025-03-27 11:00:00	Credit Card	Cancelled	2025-03-18 16:30:00
4	4	2	2025-05-02 08:45:00	2025-05-16 15:00:00	2025-05-18 11:00:00	Cash	Completed	NULL
5	5	5	2025-06-09 12:10:00	2025-06-21 15:00:00	2025-06-24 11:00:00	Credit Card	Completed	NULL
6	6	3	2025-08-14 17:05:00	2025-08-29 15:00:00	2025-08-31 11:00:00	Debit Card	Cancelled	2025-08-22 09:00:00
7	7	6	2025-10-01 11:30:00	2025-10-10 15:00:00	2025-10-13 11:00:00	Credit Card	Completed	NULL
8	8	8	2025-11-15 13:00:00	2025-12-01 15:00:00	2025-12-04 11:00:00	Credit Card	Completed	NULL
9	1	4	2026-09-15 09:00:00	2026-10-20 15:00:00	2026-10-23 11:00:00	Credit Card	Active	NULL
10	3	7	2026-09-28 15:45:00	2026-11-05 15:00:00	2026-11-07 11:00:00	Debit Card	Active	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Booking;
```

Booking table structure

Field	Type	Null	Key	Default	Extra
BookingID	int	NO	PRI	NULL	auto_increment
GuestID	int	NO	MUL	NULL	
RoomID	int	NO	MUL	NULL	
BookingDateTime	datetime	NO		NULL	
CheckInDateTime	datetime	NO		NULL	
CheckOutDateTime	datetime	NO		NULL	
PaymentMethod	varchar(45)	NO		NULL	
Status	varchar(45)	NO		NULL	
CancellationDateTime	datetime	YES		NULL	

DESC Booking;

Guest table structure

Field	Type	Null	Key	Default	Extra
GuestID	int	NO	PRI	NULL	auto_increment
FirstName	varchar(45)	NO		NULL	
LastName	varchar(45)	NO		NULL	
Email	varchar(100)	NO	UNI	NULL	
Phone	varchar(20)	YES		NULL	
IsUniversityStaff	tinyint(1)	NO		0	

DESC Guest;

Guest table data

	GuestID	FirstName	LastName	Email	Phone	IsUniversitySt...	
	1	Avery	Morgan	avery.morgan@example.com	519-555-0101	1	
	2	Noah	Patel	noah.patel@example.com	519-555-0102	0	
	3	Mia	Chen	mia.chen@example.com	519-555-0103	0	
	4	Liam	Wilson	liam.wilson@example.com	519-555-0104	1	
	5	Sophia	Brown	sophia.brown@example.com	519-555-0105	0	
	6	Ethan	Nguyen	ethan.nguyen@example.com	519-555-0106	0	
	7	Olivia	Martin	olivia.martin@example.com	519-555-0107	1	
	8	Lucas	Taylor	lucas.taylor@example.com	519-555-0108	0	
	NULL	NULL	NULL	NULL	NULL	NULL	

```
SELECT * FROM Guest;
```

RoomType table structure

	Field	Type	Null	Key	Default	Extra	
	RoomTypeID	int	NO	PRI	NULL	auto_increment	
	TypeName	varchar(45)	NO	UNI	NULL		
	Capacity	int	NO		NULL		
	NightlyPrice	decimal(10,2)	NO		NULL		

```
DESC RoomType;
```

RoomType table data

	RoomTypeID	TypeName	Capacity	NightlyPrice	
	1	Single	1	110.00	
	2	Duplex	4	185.00	
	3	Presidential Suite	2	350.00	
	NULL	NULL	NULL	NULL	

```
SELECT * FROM RoomType;
```

Room table structure

	Field	Type	Null	Key	Default	Extra
	RoomID	int	NO	PRI	NULL	auto_increment
	RoomNumber	varchar(10)	NO	UNI	NULL	
	RoomTypeID	int	NO	MUL	NULL	

```
DESC Room;
```

Room table data

	RoomID	RoomNumber	RoomTypeID
	1	101	1
	2	102	1
	3	103	1
	4	201	2
	5	202	2
	6	203	2
	7	301	3
	8	302	3
	NULL	NULL	NULL

```
SELECT * FROM Room;
```

GitHub Repository

- <https://github.com/realslimk8e/DIGT3107-DataManagement-KKQ>